CSO Lab 6: Getting to Know the 8051 MCU

Question: -- Write a program that uses a push button to start and stop a 4-bit binary up counter.

--Display the 4-bit binary up counter using the LED.

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; 4-bit Start/Stop Binary Up Counter
; Button: P0.0 (Keypad Switch A in Push Button Mode)
; Output: P1.0 - P1.3 (LED Panel)

ORG 0000h

    MOV A, #00h         ; Initialize counter to 0
    SETB F0             ; Use F0 (User Flag) as the Start/Stop state
                        ; F0 = 1 (STOPPED), F0 = 0 (RUNNING)

MAIN:
    ACALL CHECK_BUTTON  ; Check for button press to toggle F0
    JB F0, MAIN         ; If F0 is 1, stay in MAIN (Stopped)

    ; --- Counting Logic ---
    MOV P1, A           ; Output the 4-bit value to LEDs
    INC A               ; Increment count
    ANL A, #0Fh         ; Mask: Keep only bits 0, 1, 2, and 3 (4-bit count)
    SJMP MAIN           ; Repeat

; --- Toggle Logic Subroutine ---
CHECK_BUTTON:
    JB P0.0, EXIT_CB    ; If P0.0 is 1 (Not pressed), skip toggle logic

    ; Button was pressed, now wait for release (Debounce)
    ; This prevents the program from toggling too fast while holding
WAIT_RELEASE:
    JNB P0.0, WAIT_RELEASE

    CPL F0              ; Flip F0 (1 -> 0 or 0 -> 1)
EXIT_CB:
    RET

; --- Optimized Delay for Fast Blinking in Simulator ---
D1: MOV R1, #1
    DJNZ R1, $
    DJNZ R2, D1
    RET

END
```

Line: 40 Column: 1 Total: 40 INS NORM CSO LAB 6.asm ASM

Demo Video

https://drive.google.com/drive/folders/1rnfQfxfr03Xsfzd2DmIeLebdBVp3SIGy